

Dhia Eddine Arfaoui

Engineer — Machine Learning and Data Science

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Professional Summary

Machine Learning Engineer with internship experience in banking and insurance, specializing in predictive modeling, data pipelines, and API development. Skilled in Python and deep learning frameworks, with hands-on success deploying production-ready solutions for forecasting, anomaly detection, and risk analytics. A data-driven problem-solver focused on translating complex business requirements into robust, high-performance AI systems.

Experience

Data science Intern Numeryx

february 2026 – Jul 2026
Arianna, Tunisia

- Designed and implemented an AI-based system to forecast traffic and revenue for the TollXpress toll network, integrating a Data Warehouse, ML models, a REST API, and an interactive dashboard to support strategic decision-making
- Built an ETL pipeline and a star-schema Data Warehouse (PostgreSQL) consolidating toll transaction data from lanes, stations, and the central system
- Developed and benchmarked forecasting models — ARIMA, Prophet and LSTM for traffic, Linear Regression, Random Forest and XGBoost for revenue — reaching over 85% model precision
- Technologies: Python (Pandas, NumPy, Scikit-learn, Fast api ,TensorFlow), PostgreSQL , GIT , Docker, Jupyter Notebook , Risky projects .

Data and ML Intern BIAT

Jun 2025 – Jul 2025
Tunis, Tunisia

- Developed an internal API to analyze daily banking transactions and surface quantitative and qualitative indicators.
- Built SQL and Oracle data pipelines to process more than 100,000 transaction records per day for reporting and analysis.
- Performed exploratory data analysis and feature engineering to support downstream analytics and modeling tasks.
- Technologies: Python, Flask, SQL, Oracle, TensorFlow, Keras, Docker.

Data Science Intern BIAT Assurances

Jun 2024 – Jul 2024
Tunis, Tunisia

- Developed a Flask API for stochastic analysis of bonus-malus insurance classes.
- Cleaned, structured, and prepared insurance datasets for analysis and feature engineering workflows.
- Built a lightweight web interface for internal use with HTML, CSS, JavaScript, and Bootstrap.
- Technologies: Python, Flask, HTML, CSS, JavaScript, Bootstrap.

Education

Master in Artificial Intelligence, Data and Agentic (AIDA)
Université Paris Dauphine–PSL, Tunis Campus

2026 – Present
Tunis, Tunisia

- Advanced training in artificial intelligence, machine learning, data science, Big Data, and agentic and conversational systems.
- Developing intelligent algorithms, training AI models, and using cloud and Big Data tools through applied projects with strong research and industry orientation.

Engineering Degree in Industrial Engineering
ENSIT

2023 – 2026
Montfleury, Tunisia

- Multidisciplinary engineering student specializing in the intersection of financial technology , man agement , data engineering, and optimization systems. Expertise spans building machine learning pipelines for quantitative finance, implementing big data solutions with rigorous quality control, and developing high-performance algorithms using C/JAVA . Combine technical engineering skills with economic analysis and project management to deliver data-driven solutions that optimize financial processes and business outcomes

Technical Skills

- **Programming:** Python, SQL, Java, C, JavaScript
- **Machine Learning & Data Science:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost
- **Generative AI & NLP:** LangChain, FAISS, Ollama, NLTK
- **Backend & Web:** FastAPI, Flask, SQLAlchemy, Streamlit, React, Spring Boot
- **Data Engineering & Tools:** PostgreSQL, Oracle, Hadoop, HDFS, MapReduce, Docker, Git

Selected Projects

Local Document Analyzer

LangChain, Ollama, FAISS, Streamlit

- Built a local application for question-answering over PDF, TXT, and DOCX documents.
- Implemented document ingestion, text chunking, embeddings, and semantic retrieval with FAISS.
- Enabled private document analysis with local models, avoiding recurring cloud API costs.

Neural Conversational AI with LSTM

Python, Deep Learning, Seq2Seq

- Built an encoder-decoder LSTM chatbot with embeddings and attention mechanisms.
- Prepared and augmented conversational datasets, then optimized training with teacher forcing, gradient clipping, and early stopping.
- Improved response quality and multi-turn context handling with beam search decoding.

Certifications

- Associate Data Scientist, DataCamp (Issued Nov 2025, Credential ID: DSA0017594101523)
- Data Science Certification, CodeQuest (Apr 2025)